

Elizabeth Hayman

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EDUCATION and RESEARCH EXPERIENCE

DPhil in mathematical engineering, University of Oxford, Keble College; 2017 – 2021

- Four-year DPhil in the EPSRC Sustainable Approaches to Biomedical Science: Responsible and Reproducible Research Centre for Doctoral Training (DTC). Expected graduation in October 2025.
- Interdisciplinary thesis: *Generative Design of Hydrocephalus Shunts*, supervised by Professor Antoine Jerusalem and Professor Sarah Waters, in collaboration with industrial partners Lurtis Ltd. and pediatric neurosurgeon Mr Jay Jayamohan. The project combines fluid and solid finite-element models with surrogate-aided optimization algorithms to improve the design of a medical component.
- First year short research projects on adaptive cancer therapy and cerebrospinal fluid shunt design.
- Courses taken in software engineering, machine learning, computational image analysis and data science.

Masters in mathematics, University of Oxford, Worcester College; 2017 – 2021

- Achieved a first class degree (combined second and third year marks) with distinction in the 4th year masters.
- Exam grade averages: 80.5% (4th year), 89.0% (3rd year), 82.3% (2nd year), 85.9% (1st year).
- Modules taken in mathematical modelling, fluid dynamics, biological systems, probability and statistics.
- Masters dissertation *Recovery of Ancestral Graph Topologies* obtained new probabilistic results using statistical models of genetic evolution. Results extended into a first-author [journal publication](#).
- EPSRC-funded second-year summer research project on *the Thermal Profile of Lithium-ion Batteries during Short-Circuit*.

GCSEs and A level, Bartholomew's School, Eynsham; 2010 – 2027

- A Levels: A* in Maths, Further Maths, Physicals and History. A* in EPQ on *the Development of Calculus*.
- Regular awards in the British Maths Challenge and Olympiad, Physics and Astrophysics Olympiad. Top in class / school always.
- GCSEs: 11 A*s in subjects including the Sciences, English, Mathematics, French and German. A in Design.

SELECTED TEACHING and OUTREACH

College Tutor, Worcester College 2022-25, Keble College 2023-24, Lincoln College 2022.

Organized and led tutorial sessions with first- and second-year undergraduate students, covering mathematics modules across the syllabus, preparing my own extension material as needed. Marked problem sheets and mock exams and assisted with interviewing. Adapted Oxford curriculum to hold tutorials for students visiting from American institutions.

Outreach leader, Mathematics Department, 2023-24.

Held tutorial style classes for disadvantaged high school students through the UNIQ and UNIQ+ programmes. Planned and led 2-hour classes for up to 60 students aged between 14-16. Spoke about my own path and university experience on panel sessions.

RAMP Outreach Innovation Project, 2022.

Participated in organising and speaking at a public outreach event, [Maths vs Covid-19](#), aiming to demystify the use of mathematics in infectious disease modelling. Additionally created a series of YouTube videos explaining agent-based models to a general audience.

Student Mentor for Problem Solving MATters, Mathematics Department, 2018-20, 2023-4.

Guiding talented state school students through the Oxbridge application for mathematics via in-person study days and marking and providing feedback on summer work.

HONORS and AWARDS

Impact in Outreach Award, 2023, honorable mention in the outreach category at the Doctoral Training Centre 21st Anniversary awards.

RAMP Outreach Innovation Award, 2022, awarded £5000 group funding to organise public engagement events.

EPSRC studentship, 2021, awarded funding for four years of DPhil study.

Oxford University Junior Mathematical Prize, 2020, outstanding performance in third year exams. One of two such prizes awarded.

EPSRC Poster Presentation Award, 2019, award for the best summer project poster presentation across multiple disciplines.

EPSRC Vacation Placement Award, 2019, awarded funding for an 8-week undergraduate research project modelling short-circuits in lithium-ion batteries.

Oxford Academic Scholarship, 2017, 2018, 2019 and 2020 award for first class performance during exams.

CONFERENCES and SUMMER SCHOOLS

- 19th European Mechanics of Materials Conference, Madrid, 2024. Talk given.
- SIG Discovery Day - Quantitative Research & Systematic Trading, Dublin, 2024.
- IBME Bellhouse BioEngenuity, Oxford, 2024. Poster presentation.
- Spring into Quant Finance, G-research insights week, Nice, 2023. Participated in promotional [video](#).
- Flagship Retreat for Women in Applied Mathematics, Edinburgh, 2023. Poster presentation.
- Research Students' Conference in Population Genetics, Warwick, 2022. Talk given.
- Summer School for Mathematical Ecology and Evolution, Helsinki, 2022.
- The Role of Uncertainty in Mathematical Modelling of Pandemics, Cambridge 2022.

SOFTWARE SKILLS

- Daily use of python including the scipy, pytorch and matplotlib subpackages for DPhil research.
- Familiarity with C++ and git. Contributor to my group's open-source in-house solids modelling software [MuPhiSim](#).
- Regular user of fluid simulation and visualisation softwares; OpenFOAM and paraview.
- Year-long DTC group software project creating an open-source epidemiological agent-based model [epiabm](#). Software and preliminary results [published](#).
- DTC courses in responsible coding, focusing on test-driven development and reproducibility. Course work assessed via Kaggle competitions.

EXTRACURRICULAR INTERESTS

I enjoy ice dancing (NISA level 8, university society treasurer), mountain biking (university club captain, three Oxford half-blues, women's gold at Varsity 2024) and triathlon. I study piano (grade 8), viola (grade 5) and organ. I am a member of the college orchestra and enjoy attending classical concerts and Gilbert and Sullivan operettas. I am always keen to explore new cultures, most recently spending three weeks solo-travelling Peru.